

Notes: Costinot, Donaldson, and Komunjer (RES, 2012)

What Goods Do Countries Trade? A Quantitative Exploration of Ricardo's Ideas

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1 Big picture

- **Question.** The paper asks whether the central Ricardian prediction is visible in modern data: do countries export relatively more in industries in which they are relatively more productive?
- **The core problem.** Classical Ricardian theory is elegant but difficult to test directly because productivity in non-produced goods is unobserved, multi-country comparative advantage is hard to summarize, and earlier empirical work often used ad hoc export-performance measures.
- **Theoretical move.** The paper builds a multi-country, multi-industry Ricardian model with intra-industry heterogeneity in the spirit of Eaton and Kortum (2002). Each industry contains many varieties, and country-industry productivity has both a fundamental component and an idiosyncratic variety-level component.
- **Empirical move.** The model tells the authors exactly what the dependent variable should be: log bilateral exports by exporter, importer, and industry, differenced across exporters and industries, and corrected for trade-driven selection using domestic expenditure shares.
- **Productivity measurement.** Relative productivity is measured mainly through inverse producer prices from the GGDC Productivity Level Database, because in a Ricardian model relative producer prices reveal relative unit costs.
- **Main cross-sectional result.** The preferred IV estimate implies that the elasticity of corrected bilateral exports with respect to observed productivity is about 6.53. This is positive, statistically significant, and close to existing estimates of the Eaton-Kortum trade elasticity.
- **Selection correction.** Correcting exports for openness is quantitatively important. Without the correction, the IV estimate rises to about 11.10, which the model interprets as upward bias caused by the gap between observed and fundamental productivity.
- **Counterfactual question.** The paper then asks what would happen if, for any pair of exporters, relative productivity differences across industries were removed - that is, if industry-level Ricardian comparative advantage disappeared.
- **Main welfare result.** Removing industry-level Ricardian comparative advantage lowers the total gains from trade by only about 5.3% on average across the sample countries, although there is substantial heterogeneity across countries.
- **Economic takeaway.** Ricardian productivity differences clearly help explain the commodity pattern of trade, but industry-level Ricardian forces account for only a modest share of aggregate gains from trade once heterogeneous trade costs and heterogeneous preferences are allowed.

2 Incremental contribution of the paper

- **From informal Ricardian tests to structural Ricardian testing.** Earlier empirical Ricardian work often compared exports and productivity using measures such as total exports or bilateral net exports. This paper derives the export object directly from a structural model.
- **A theory-consistent dependent variable.** The paper shows that the correct empirical object is not simply exports, but *corrected bilateral exports*: exports divided by the domestic expenditure share in the exporting country and industry. This correction handles trade-driven selection into domestic production.

- **Observed versus fundamental productivity.** A key contribution is the transparent distinction between fundamental productivity, which is the model primitive, and observed productivity, which is conditional on the varieties a country actually produces.
- **A new revealed-productivity measure.** The paper provides a theoretically grounded alternative to Balassa’s revealed comparative advantage index by estimating exporter-industry fixed effects from bilateral trade flows.
- **Bridge between old Ricardo and Eaton-Kortum.** The paper uses Eaton-Kortum micro-foundations to recover a clear multi-country, multi-industry version of Ricardo’s prediction: pairwise productivity differences predict pairwise relative export performance.
- **Cross-sectional and counterfactual unity.** The same structural parameter θ estimated from cross-sectional regressions is used to discipline counterfactual welfare analysis.
- **Quantification of Ricardian welfare importance.** The paper goes beyond asking whether Ricardian comparative advantage exists. It asks how much welfare would change if that source of comparative advantage were removed.
- **Role of demand and trade costs.** The paper shows that heterogeneous preferences and heterogeneous trade costs can offset productivity-based comparative advantage, substantially lowering the welfare importance of industry-level Ricardian forces.
- **Practical contribution for empirical trade.** The paper gives a replicable template for using bilateral trade data, productivity data, fixed effects, IV, and structural counterfactuals to quantify comparative advantage.

3 Data and measurement

3.1 Trade-flow data

- The main trade data are from the OECD STAN Bilateral Trade Database, edition 2008.
- The empirical sample uses the year 1997, chosen because it is the intersection of available high-quality trade-flow data and internationally comparable productivity data.
- The sample contains 21 countries and 13 manufacturing industries at roughly the two-digit ISIC level.
- The key trade-flow variable is x_{ij}^k , the value of bilateral exports from exporter i to importer j in industry k .
- The paper also uses import penetration ratios from OECD STAN to construct the domestic expenditure share π_{ii}^k , which is needed for the selection correction.

Interpretation: The empirical design requires a square country-by-country trade matrix because the counterfactual analysis needs bilateral trade shares for all exporter-importer pairs in the sample.

3.2 Productivity data

- Productivity is measured using producer price data from the GGDC Productivity Level Database.

- In the Ricardian model, lower producer prices reflect lower unit costs and therefore higher productivity.
- The paper takes the inverse of the producer price index as the empirical counterpart of observed productivity $\tilde{z}_i^k$.
- Productivity levels are normalized so that the U.S. has productivity equal to one within each industry and the food industry has productivity equal to one within each country.
- Alternative productivity measures include a dual TFP measure, real gross output per worker, and a primal TFP measure.

Interpretation: Producer prices are useful because the model is ultimately about comparative costs. The main limitation is measurement error, which motivates the IV strategy.

3.3 Observed productivity, fundamental productivity, and openness

- Fundamental productivity z_i^k is a model primitive: the underlying productivity of country i in industry k .
- Observed productivity $\tilde{z}_i^k$ is conditional on the varieties actually produced in country i .
- Trade changes what is produced domestically. Countries with low fundamental productivity in an industry may only produce varieties for which they receive unusually good idiosyncratic draws.
- Therefore, observed productivity can be higher than fundamental productivity for more open countries because low-productivity varieties are selected out of domestic production.
- The domestic expenditure share π_{ii}^k corrects for this trade-driven selection.

Interpretation: The selection correction is the paper’s central measurement insight. Without it, observed productivity differences understate fundamental productivity differences, biasing the export-productivity elasticity.

3.4 Revealed productivity for counterfactuals

- For counterfactual analysis, the authors do not rely directly on measured producer-price productivity because it contains substantial measurement error.
- Instead, they estimate exporter-industry fixed effects from bilateral trade flows and interpret them as revealed fundamental productivity.
- This revealed-productivity approach is structurally related to Balassa’s revealed comparative advantage, but it uses bilateral flows and fixed effects to separate productivity from importer demand and trade-cost components.

Interpretation: The revealed-productivity measure makes the welfare counterfactual less sensitive to noise in producer prices, while remaining disciplined by the Ricardian model.

4 Models and empirical specifications

4.1 Ricardian model with intra-industry heterogeneity

The model has multiple countries $i = 1, \dots, I$, multiple industries $k = 1, \dots, K$, and one factor of production, labour. Each industry contains infinitely many varieties ω . Labour productivity for variety ω in country i and industry k is drawn from a Frechet distribution:

$$F_i^k(z) = \exp \left[- \left(\frac{z}{z_i^k} \right)^{-\theta} \right], \quad z \geq 0. \quad (1)$$

- z_i^k is fundamental productivity for country i in industry k .
- θ governs intra-industry heterogeneity and the elasticity of trade flows with respect to productivity.
- A higher z_i^k means that country i has better productivity draws in industry k .
- Iceberg trade costs d_{ij}^k make it costly to ship goods from exporter i to importer j .

Interpretation: The model preserves the Ricardian idea that productivity determines comparative advantage, but it avoids complete specialization by allowing idiosyncratic productivity variation within each industry.

4.2 Bilateral exports at the industry level

Under perfect competition and CES demand over varieties, bilateral exports in industry k can be written as

$$x_{ij}^k = \frac{(w_i d_{ij}^k / z_i^k)^{-\theta}}{\sum_{i'=1}^I (w_{i'} d_{i'j}^k / z_{i'}^k)^{-\theta}} \alpha_j^k w_j L_j. \quad (2)$$

A key implication is the difference-in-differences relationship:

$$\ln \left(\frac{x_{ij}^k x_{i'j}^{k'}}{x_{ij}^{k'} x_{i'j}^k} \right) = \theta \ln \left(\frac{z_i^k z_{i'}^{k'}}{z_i^{k'} z_{i'}^k} \right) - \theta \ln \left(\frac{d_{ij}^k d_{i'j}^{k'}}{d_{ij}^{k'} d_{i'j}^k} \right). \quad (3)$$

- The first log difference controls for exporter wages and importer income.
- The second log difference controls for importer-sector expenditure shares.
- The remaining productivity term is the Ricardian comparative-advantage term.

Interpretation: Relative productivity across two exporters and two industries predicts relative exports, conditional on relative trade costs.

4.3 Observed productivity and corrected exports

Observed productivity differs from fundamental productivity because only domestically produced varieties are observed. The model implies

$$\frac{z_i^k}{z_{i'}^k} = \left(\frac{\tilde{z}_i^k}{\tilde{z}_{i'}^k} \right) \left(\frac{\pi_{ii}^k}{\pi_{i'i'}^k} \right)^{1/\theta}. \quad (4)$$

Define corrected exports as

$$\tilde{x}_{ij}^k = \frac{x_{ij}^k}{\pi_{ii}^k}. \quad (5)$$

Then the cross-sectional prediction becomes

$$\ln \left(\frac{\tilde{x}_{ij}^k \tilde{x}_{i'j}^{k'}}{\tilde{x}_{ij}^{k'} \tilde{x}_{i'j}^k} \right) = \theta \ln \left(\frac{\tilde{z}_i^k \tilde{z}_{i'}^{k'}}{\tilde{z}_i^{k'} \tilde{z}_{i'}^k} \right) - \theta \ln \left(\frac{d_{ij}^k d_{i'j}^{k'}}{d_{ij}^{k'} d_{i'j}^k} \right). \quad (6)$$

Interpretation: Corrected exports are the empirical object that makes observed productivity usable for Ricardian testing.

4.4 Cross-sectional regression

The empirical regression corresponding to the model is

$$\ln \tilde{x}_{ij}^k = \delta_{ij} + \delta_j^k + \theta \ln \tilde{z}_i^k + \varepsilon_{ij}^k. \quad (7)$$

- δ_{ij} are exporter-importer fixed effects.
- δ_j^k are importer-industry fixed effects.
- θ is the elasticity of corrected exports with respect to observed productivity.
- The error term captures relative trade costs and measurement error.
- The preferred specification instruments productivity with country-industry R&D expenditure.

Interpretation: The fixed effects convert the model's pairwise difference-in-difference prediction into a regression using all bilateral exporter-importer-industry observations.

4.5 Counterfactual removal of Ricardian comparative advantage

The paper asks what would happen if, relative to a reference country, all other countries had the same relative productivity profile across industries. Absolute productivity is adjusted so that relative wages are held fixed, isolating the effect of removing industry-level Ricardian comparative advantage.

For bilateral trade flows, the proportional change satisfies

$$\hat{x}_{ij}^k = \frac{(z_i^k/Z_i)^{-\theta}}{\sum_{i'=1}^I \pi_{i'j}^k (z_{i'}^k/Z_{i'})^{-\theta}}. \quad (8)$$

For welfare in the reference country i_0 , the proportional change satisfies

$$\hat{W}_{i_0} = \prod_{k=1}^K \left[\sum_{i=1}^I \pi_{ii_0}^k \left(\frac{z_i^k}{z_{i_0}^k Z_i} \right)^{-\theta} \right]^{\alpha_{i_0}^k / \theta}. \quad (9)$$

Interpretation: The counterfactual compares the observed world to one in which countries no longer have different relative productivity rankings across industries.

5 Identification and empirical design

5.1 What the design identifies

- The cross-sectional regressions identify whether relative productivity differences are associated with relative export performance across countries and industries.
- The coefficient θ is interpreted structurally as the trade elasticity associated with productivity differences in the Ricardian model.
- The counterfactuals identify model-implied changes in trade and welfare when industry-level productivity differences are removed.

Interpretation: The empirical design is structural rather than purely reduced-form. The regression coefficient is meaningful because the model determines the dependent variable, the fixed effects, and the interpretation of θ .

5.2 Endogeneity and instrumental variables

- OLS may suffer from attenuation bias because internationally comparable producer prices are difficult to measure.
- OLS may also suffer from simultaneity if exporting affects productivity through agglomeration or learning channels.
- The preferred IV specification instruments productivity with country-industry R&D expenditure.
- The identifying assumption is that R&D affects relative exports through productivity rather than through other omitted determinants of trade.

Interpretation: The large increase from OLS to IV estimates suggests that measurement error in productivity is important.

5.3 Trade costs and omitted variables

- The error term contains relative trade costs that vary across exporter, importer, and industry.
- If relative productivity is correlated with relative trade protection, estimates may be biased.
- The paper addresses this concern by estimating the model within EU trade pairs, where trade barriers are lower and less likely to vary with protectionist policy.
- The estimates remain positive and significant, supporting the baseline interpretation.

Interpretation: The EU sample is an important robustness check because it reduces concerns that productivity is merely proxying for trade-policy differences.

5.4 Productivity-measure robustness

- The main productivity measure is based on inverse producer prices.
- The authors also use a dual TFP measure that removes variation in factor prices and factor shares.
- The estimate remains very close to the baseline: 6.70 versus 6.53.
- Alternative primal productivity measures give lower coefficients, which the paper interprets as consistent with additional measurement error.

Interpretation: The similarity between inverse producer-price and dual-TFP estimates supports the Ricardian reading of the results rather than a pure Heckscher-Ohlin factor-price interpretation.

5.5 Counterfactual interpretation

- The counterfactual removes relative productivity differences across industries, but it does not remove all Ricardian forces because there remains intra-industry heterogeneity.
- The adjustment of absolute productivity keeps wages fixed, which neutralizes terms-of-trade effects for the reference country.
- The welfare results should be read as model-implied magnitudes rather than direct empirical treatment effects.

Interpretation: The counterfactual answers a very precise question: how important is industry-level Ricardian comparative advantage, holding the rest of the model structure fixed?

6 Tables and figures

6.1 Table 1: Data set description

Read:

- Table 1 describes the core sample: OECD STAN bilateral trade data and GGDC productivity data for 1997.

- The sample includes 21 countries and 13 manufacturing industries.
- The industry list ranges from food, textiles, wood, paper, fuel, chemicals, plastics, minerals, and metals to machinery, electrical equipment, transport equipment, and miscellaneous manufacturing.
- The table makes clear that the paper is constrained by the availability of comparable trade-flow and productivity data.

Economic message: The sample is designed to balance cross-country coverage with comparable productivity measurement. The empirical test is therefore conducted at a relatively aggregated two-digit manufacturing-industry level.

TABLE 1
Data set description

Sources	OECD STAN Bilateral Trade Database (Edition 2008); GGDC Productivity Level Database: International Comparisons of Output, Inputs and Productivity at the Industry Level (Inklaar and Timmer, 2008).	
Year	1997	
Countries	Australia, Belgium–Luxembourg (aggregated into one country unit to enable a merge with the productivity data), Czech Republic, Denmark, Spain, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Japan, Korea, the Netherlands, Poland, Portugal, Slovakia, Sweden, U.K., and U.S.	
Industry	STAN description	ISIC Rev. 3.1 Code
Food	Food products, beverages, and tobacco	15–16
Textiles	Textiles, textile products, leather, and footwear	17–19
Wood	Wood and products of wood and cork	20
Paper	Pulp, paper, paper products, printing and publishing	21–22
Fuel	Coke, refined petroleum products, and nuclear fuel	23
Chemicals	Chemicals and chemical products	24
Plastic	Rubber and plastics products	25
Minerals	Other non-metallic mineral products	26
Metals	Basic metals and fabricated metal products, except machinery and equipment	27–28
Machinery	Machinery and equipment not elsewhere classified	29
Electrical	Electrical and optical equipment	30–33
Transport	Transport equipment	34–35
Misc. Manuf	Manufacturing not elsewhere classified	36–37

TABLE 2
Relative productivity levels, by country and industry

	Food	Textiles	Wood	Paper	Fuel	Chemicals	Plastics	Minerals	Metals	Machinery	Electrical	Transport	Misc.Manuf
AUS	1	0.77	0.89	0.85	1.21	0.84	0.81	1.00	0.84	0.86	0.82	0.74	0.84
BEL	1	0.84	0.84	1.03	1.70	1.26	1.71	1.49	1.07	0.93	0.86	0.71	1.11
CZE	1	0.80	1.16	0.72	0.66	1.44	1.65	1.88	1.05	0.97	0.93	0.78	0.85
DEU	1	0.75	0.84	0.63	0.64	1.02	1.01	0.98	0.85	0.85	0.63	0.51	1.35
ESP	1	0.92	1.47	0.91	0.70	1.34	1.38	1.58	1.08	0.99	1.08	1.01	1.15
FIN	1	0.82	1.20	1.18	0.86	1.69	1.37	1.30	1.18	1.13	0.90	0.65	1.47
FRA	1	0.76	1.22	0.93	1.11	1.24	1.64	1.28	1.08	1.13	0.93	0.84	0.92
GER	1	0.67	0.91	0.89	0.94	1.03	1.23	1.20	0.91	0.92	0.80	0.72	0.90
GRC	1	1.07	1.65	1.57	1.13	1.87	1.21	2.19	1.51	1.54	1.26	0.83	1.36
HUN	1	0.80	1.05	1.08	0.52	0.98	1.32	1.33	0.91	1.52	0.83	0.49	1.14
IRE	1	0.93	1.03	0.81	1.15	1.49	1.55	1.19	1.11	0.92	0.74	0.62	1.11
ITA	1	0.79	1.02	0.68	0.65	1.26	1.48	1.63	1.17	1.02	0.98	1.14	0.94
JPN	1	1.88	1.37	1.69	1.06	1.84	1.69	1.71	1.85	2.05	1.96	1.96	1.33
KOR	1	1.23	1.56	1.40	1.32	2.74	1.97	2.22	1.50	1.64	1.69	1.72	1.94
NED	1	0.85	0.54	0.76	0.65	1.25	1.26	1.11	1.00	0.71	0.75	0.61	0.74
POL	1	1.15	1.12	0.92	1.07	1.34	2.21	1.59	0.91	0.80	0.60	0.68	0.64
PTL	1	1.18	1.32	1.08	0.62	1.26	2.32	1.66	1.42	1.01	1.01	0.59	0.91
SLK	1	0.63	0.82	0.91	0.49	1.04	1.07	1.45	0.71	0.63	0.72	0.65	0.89
SWE	1	0.68	1.18	0.92	0.78	1.22	1.18	0.92	1.14	0.98	0.89	0.67	1.09
U.K.	1	0.92	0.69	1.23	1.27	1.20	1.74	1.24	1.02	1.29	0.97	0.79	1.80
U.S.	1	1	1	1	1	1	1	1	1	1	1	1	1

Table 2: Relative productivity levels, by country and industry.

Table 1: Data set description.

6.2 Table 2: Relative productivity levels

Read:

- Table 2 reports relative productivity based on inverse producer prices.
- The normalization sets the U.S. equal to one within every industry and food equal to one within every country.
- The table shows substantial cross-country and cross-industry productivity dispersion.
- For example, Japan and Korea often have high relative productivity in machinery, electrical equipment, transport, and several manufacturing sectors.
- These observed productivity measures are later used in the cross-sectional export-productivity regressions.

Economic message: The table supplies the central explanatory variable: relative productivity. The large dispersion across countries and industries provides the variation needed to test the Ricardian prediction.

6.3 Tables 3 and 4: Baseline cross-sectional results and alternative productivity measures

Read:

- Table 3 estimates the main cross-sectional regression. The preferred IV specification with corrected exports gives $\theta = 6.534$, highly significant.
- Comparing columns with and without corrected exports shows that failing to adjust for trade-driven selection materially changes the estimate.
- Table 4 shows robustness to alternative productivity measures. The dual-TFP measure produces a very similar coefficient, 6.704.
- Real gross output per worker and primal TFP give smaller coefficients, consistent with greater measurement error in those productivity proxies.

Economic message: The export-productivity relationship is positive and quantitatively strong. The similarity between producer-price and dual-TFP estimates supports the view that the regression is capturing productivity-based comparative advantage.

TABLE 3
Cross-sectional results—baseline

Dependent variable	log (corrected exports)	log (exports)	log (corrected exports)	log (exports)
	(1)	(2)	(3)	(4)
log (productivity based on producer prices)	1.123*** (0.0994)	1.361*** (0.103)	6.534*** (0.708)	11.10*** (0.981)
Estimation method	OLS	OLS	IV	IV
Exporter $\times$ importer fixed effects	YES	YES	YES	YES
Industry $\times$ importer fixed effects	YES	YES	YES	YES
Observations	5652	5652	5576	5576
R^2	0.856	0.844	0.747	0.460

Notes: Regressions estimating equation (18) using data from 21 countries and 13 manufacturing sectors (listed in Table 1) in 1997. “Exports” is the value of bilateral exports from the exporting country to the importing country in a given industry. “Corrected exports” is “exports” divided by the share of the exporting country’s total expenditure in the given industry that is sourced domestically (equal to one minus the country and industry’s IPR). “Productivity based on producer prices” is the inverse of the average producer price in an exporter–industry. Columns (3) and (4) use the log of 1997 R&D expenditure as an instrument for productivity. Data sources and construction are described in full in Section 4.1. Heteroskedasticity-robust standard errors are reported in parentheses. ***Statistically significantly different from zero at the 1% level.

(a) Table 3: Baseline cross-sectional results.

TABLE 4
Cross-sectional results—alternative productivity measures

Dependent variable in all regressions: corrected exports				
	(1)	(2)	(3)	(4)
log (productivity based on producer prices)	6.534*** (0.708)			
log (productivity based on producer prices, dual TFP measure)		6.704*** (0.874)		
log (productivity based on real gross output per worker)			2.725*** (0.234)	
log (productivity based on real gross output, primal TFP measure)				4.316*** (0.504)
Observations	5576	5576	5576	4541
R^2	0.747	0.587	0.839	0.835

Notes: Regressions are based on estimating equation (18) using data from 21 countries and 13 manufacturing sectors (listed in Table 1) in 1997. All regressions estimated use the log of 1997 R&D expenditure as an instrument for the productivity measure and with exporter–times–importer and industry–times–importer fixed effects included. “Corrected exports” is the value of bilateral exports divided by the share of the exporting country’s total expenditure in the given industry that is sourced domestically (equal to one minus the country and industry’s IPR). “Productivity based on producer prices” is the inverse of the average producer price in an exporter–industry. “Productivity based on producer prices, dual TFP measure” is the inverse of the average producer price divided by an income share–weighted composite of producer input prices (wages, capital rental rates, and intermediate inputs), as defined in footnote 24. “Productivity based on real gross output per worker” is a nominal measure of gross output, deflated by average gross output producer price indices. “Productivity based on real gross output, primal TFP measure” is the nominal value of gross output deflated by a producer gross output price index, divided by an income share–weighted composite of input values deflated by input price indices, as defined in footnote 26. Data sources and construction are described in full in Section 4.1. Heteroskedasticity-robust standard errors are reported in parentheses. ***Statistically significantly different from zero

(b) Table 4: Alternative productivity measures.

6.4 Table 5: Alternative samples

Read:

- Table 5 examines whether the preferred estimate is robust to different samples.
- Restricting to EU member-country trade pairs gives $\theta = 4.621$, still positive and significant.
- Splitting by producer-price data quality gives estimates of 5.820 below the median and 8.057 above the median.
- These estimates remain in the same broad range as the baseline and as prior estimates of the Eaton-Kortum elasticity.

Economic message: The main finding is not driven by a particular sample or by low-quality price data. The robustness checks strengthen the structural interpretation of the baseline estimate.

TABLE 5
Cross-sectional results—alternative samples

	Dependent variable in all regressions: corrected exports			
	Entire sample	EU member countries only	Producer price data below median quality	Producer price data above median quality
	(1)	(2)	(3)	(4)
log (productivity based on producer prices)	6.534*** (0.708)	4.621*** (0.585)	5.820*** (1.106)	8.057*** (1.267)
Observations	5576	2162	2795	2781
R^2	0.747	0.808	0.820	0.688

Notes: Regressions are based on estimating equation (18) using data from 21 countries and 13 manufacturing sectors (listed in Table 1) in 1997. All regressions instrument for productivity using log R&D expenditures in 1997 and include exporter-times-importer fixed effects and industry-times-importer fixed effects. “Corrected exports” is the value of bilateral exports divided by the share of the exporting country’s absorption in the given industry that is sourced domestically (equal to one minus the country and industry’s IPR). “Productivity based on producer prices” is the inverse of the average producer price in an exporter-industry. “Producer price data quality” is based on the percentage of underlying price data observations within the exporter-industry that were obtained (in the creation of the GGDC data set) using producer price data rather than expenditure price data. Data sources and construction are described in full in Section 4.1. Heteroskedasticity-robust standard errors are reported in parentheses. ***Statistically significantly different from zero at the 1% level.

TABLE 6
Revealed relative productivity levels, by country and industry

	Food	Textiles	Wood	Paper	Fuel	Chemicals	Plastic	Minerals	Metals	Machinery	Electrical	Transport	Misc. Manuf
AUS	1	0.91	0.73	0.63	0.74	0.75	0.74	0.69	0.88	0.71	0.71	0.68	0.77
BEL	1	1.16	0.87	0.91	1.03	1.06	1.07	1.07	1.12	0.89	0.80	0.89	0.98
CZE	1	1.37	1.16	1.07	0.93	1.05	1.21	1.56	1.41	1.13	0.90	1.10	1.23
DEN	1	0.90	0.85	0.77	0.65	0.77	0.90	0.82	0.89	0.83	0.72	0.63	0.86
ESP	1	1.14	0.86	0.86	0.81	0.87	1.03	1.16	1.08	0.83	0.78	0.96	0.94
FIN	1	1.09	1.54	1.70	0.97	1.04	1.17	1.19	1.46	1.17	1.19	0.90	1.09
FRA	1	1.08	0.83	0.91	0.97	0.97	1.00	1.06	1.07	0.88	0.85	0.90	0.90
GER	1	1.20	0.96	1.05	0.98	1.06	1.20	1.17	1.26	1.09	0.97	1.08	1.04
GRC	1	1.16	0.59	0.74	0.86	0.72	0.83	1.00	0.97	0.65	0.59	0.50	0.75
HUN	1	1.13	0.89	0.84	0.79	0.89	0.98	1.06	1.06	0.82	0.87	0.84	0.82
IRE	1	0.83	0.64	1.08	0.57	0.98	0.85	0.76	0.82	0.72	0.92	0.55	0.82
ITA	1	1.44	0.92	0.96	0.82	0.97	1.16	1.29	1.21	1.10	0.86	0.92	1.22
JPN	1	1.51	0.87	1.31	1.10	1.67	1.84	1.75	1.80	1.80	1.81	1.93	1.72
KOR	1	1.69	0.84	0.92	0.87	1.20	1.46	1.04	1.36	1.21	1.32	1.42	1.34
NED	1	0.96	0.68	0.87	0.35	0.89	0.92	0.87	0.96	0.77	0.81	0.74	0.84
POL	1	1.18	1.06	0.91	0.99	0.92	0.99	1.18	1.26	0.86	0.83	0.84	1.06
PTL	1	1.55	1.32	1.03	0.79	0.87	1.05	1.31	1.05	0.89	0.90	0.95	0.94
SLK	1	1.33	1.24	1.14	0.77	1.13	1.32	1.48	1.63	1.09	0.93	0.91	1.16
SWE	1	1.12	1.37	1.39	1.16	1.08	1.34	1.15	1.40	1.14	1.09	1.04	1.12
UK	1	1.16	0.75	1.03	1.09	1.01	1.08	1.09	1.14	0.96	0.95	0.93	0.99
U.S.	1	1	1	1	1	1	1	1	1	1	1	1	1

Table 5: Cross-sectional results, alternative samples.

Table 6: Revealed relative productivity levels.

Table 1: Data set description.

6.5 Table 6: Revealed relative productivity

Read:

- Table 6 reports revealed productivity inferred from exporter-industry fixed effects in bilateral trade-flow regressions.
- The normalization is the same as in Table 2: U.S. equal to one by industry and food equal to one by country.
- The table is used for the counterfactual analysis because it relies on trade-implied productivity rather than noisy producer-price productivity.
- The revealed measures differ from Table 2 because they absorb information from trade flows after controlling for importer-industry and exporter-importer fixed effects.

Economic message: Table 6 is a model-based alternative to revealed comparative advantage. It converts bilateral trade patterns into structural productivity measures for welfare counterfactuals.

6.6 Tables 7 and 8: Counterfactual results

Read:

- Table 7 reports the baseline counterfactual in which industry-level Ricardian comparative advantage is removed country by country.
- The average welfare effect is small: a 0.49% fall in welfare, equal to about 5.32% of the total gains from trade.
- The country-level effects are heterogeneous. Australia and Greece lose a large share of their gains from trade, while some countries experience small gains.
- Table 8 shows robustness counterfactuals. Removing preference differences and trade-cost heterogeneity raises the welfare importance of Ricardian comparative advantage.

- When both preference differences and trade-cost heterogeneity are removed, the average welfare effect rises to 27.31% of total gains from trade.

Economic message: Industry-level Ricardian forces matter, but their aggregate welfare importance is muted by heterogeneous preferences and trade costs. The textbook model overstates the welfare importance of productivity differences if it ignores these other sources of trade patterns.

TABLE 7
Counterfactual results—baseline

Reference country	Outcome variable of interest			
	% change in total exports	Change in index of interindustry trade	% change in welfare	% change in welfare relative to the total gains from trade
	(1)	(2)	(3)	(4)
Australia	18.52	24.57	-2.90	-39.11
Belgium and Luxembourg	-1.76	4.12	0.71	2.64
Czech Republic	3.91	5.62	-0.12	-1.26
Denmark	0.60	-2.64	-0.40	-2.18
Spain	3.68	-3.89	-0.46	-7.08
Finland	-5.62	3.44	0.14	1.65
France	0.80	-0.49	-0.20	-3.09
Germany	-2.10	-8.46	0.14	2.22
Greece	26.35	-11.23	-4.37	-40.47
Hungary	1.70	-5.28	-0.25	-1.62
Ireland	-5.48	-4.31	0.20	0.74
Italy	-4.76	-9.85	0.14	2.78
Japan	-6.12	-24.75	0.35	24.48
Korea	2.68	-10.15	-0.44	-9.60
Netherlands	1.95	-0.94	-0.64	-2.81
Poland	12.33	-22.35	-1.68	-23.09
Portugal	8.44	-13.62	-0.92	-9.12
Slovakia	2.33	14.11	0.82	4.64
Sweden	-2.98	3.03	0.34	3.30
U.K.	3.45	-4.04	-0.26	-2.94
U.S.	3.82	-3.83	-0.42	-11.71
World average	2.94	-5.72	-0.49	-5.32

Notes: Results from counterfactual calculations in which, one at a time for each country listed (the “reference country”), every other country in the world is given the reference country’s relative productivity levels across industries, while adjusting each country’s absolute productivity levels in such a way as to hold nominal wages fixed around the world (so as to neutralize terms-of-trade effects). The methodology follows Lemma 4 and Theorem 5. Column (1) reports the resulting proportional change in the total volume of the reference country’s exports. Column (2) reports the resulting change in the reference country’s index of inter-industry trade (given in Section 5.2); this index takes the value of 100 in the case of pure inter-industry trade and zero in the case of pure intra-industry trade. Column (3) reports the resulting proportional change in welfare (equal to real income spent on manufacturing) for the reference country; a negative number indicates a welfare loss. And Column (4) reports this change in welfare as a percentage of the total gains from trade; a negative number indicates a welfare loss (and a value of 100 means that this loss is equal to that of moving to

TABLE 8
Counterfactual results—robustness

Counterfactual scenario	Outcome variable of interest			
	% change in total exports	Change in index of inter-industry trade	% change in welfare	% change in welfare relative to the total gains from trade
	(1)	(2)	(3)	(4)
Baseline (as in Table 7)	2.94	-5.72	-0.49	-5.32
No preference differences	2.63	-5.29	-1.59	-11.74
No trade costs	3.46	-11.17	-0.74	-9.92
No preference differences or trade costs	3.46	-11.17	-2.83	-27.31

Notes: Results from counterfactual calculations in which, one at a time for each country listed (the “reference country”), every other country in the world is given the reference country’s relative productivity levels across industries, while adjusting each country’s absolute productivity levels in such a way as to hold nominal wages fixed around the world (so as to neutralize terms-of-trade effects). The methodology follows Lemma 4 and Theorem 5. Column (1) reports the resulting proportional change in the total volume of the reference country’s exports. Column (2) reports the resulting change in the reference country’s index of inter-industry trade (given in Section 5.2); this index takes the value of 100 in the case of pure inter-industry trade and zero in the case of pure intra-industry trade. Column (3) reports the resulting proportional change in welfare (equal to real income spent on manufacturing) for the reference country; a negative number indicates a welfare loss. And Column (4) reports this change in welfare as a percentage of the total gains from trade; a negative number indicates a loss (and a value of 100 means that this loss is equal to that of moving to autarky). Unweighted world average effects (averaging across all reference countries) of these counterfactuals are reported. The methodology follows Lemma 4 and Theorem 5. The “No preference differences” scenario calculates world expenditure

(a) Table 7: Baseline counterfactual results.

(b) Table 8: Counterfactual robustness.

7 Short summary

- The paper develops a multi-country, multi-industry Ricardian model with Eaton-Kortum-style intra-industry heterogeneity.
- It uses the model to derive a theory-consistent Ricardian test using bilateral exports, observed productivity, exporter-importer fixed effects, importer-industry fixed effects, and an openness correction.
- The correction for trade-driven selection is central: corrected exports are defined as exports divided by the domestic expenditure share in the exporter-industry.
- The data combine OECD STAN bilateral trade flows with GGDC producer-price productivity measures for 21 countries and 13 manufacturing industries in 1997.
- The preferred IV estimate is $\theta = 6.53$, meaning that corrected exports respond strongly to productivity differences.
- Robustness checks using alternative productivity measures and alternative samples support the main estimate.

- The paper constructs revealed productivity measures from exporter-industry fixed effects as a structural alternative to Balassa-style revealed comparative advantage.
- Counterfactuals show that removing industry-level Ricardian comparative advantage reduces total gains from trade by only about 5.3% on average.
- Heterogeneous preferences and heterogeneous trade costs explain why the welfare role of industry-level Ricardian forces is modest in the baseline.
- The paper’s main contribution is to make Ricardo empirically testable and quantitatively interpretable in a modern structural trade framework.

8 Conclusion

8.1 Core conclusion

Costinot, Donaldson, and Komunjer show that the Ricardian model has clear empirical content once it is embedded in a multi-country, multi-industry Eaton-Kortum-style framework. Countries export relatively more in industries in which they are relatively more productive, and this relationship is quantitatively strong after correcting for trade-driven selection. The preferred estimate of the productivity-to-exports elasticity, $\theta = 6.53$, confirms that productivity differences are an important determinant of the commodity pattern of trade.

8.2 Economic intuition

The economic intuition is simple. Higher productivity lowers unit costs and makes a country more competitive in an industry. However, because countries trade, observed productivity is not the same as fundamental productivity. More open countries may stop producing low-productivity varieties and only produce the varieties where their idiosyncratic draws are favourable. Therefore, a simple comparison of observed productivity and exports is biased. The paper’s corrected-export measure removes this selection wedge and recovers the Ricardian relationship more cleanly.

8.3 Incremental contribution revisited

The paper advances the trade literature in three main ways. First, it gives a structural foundation to empirical Ricardian regressions that had previously been viewed as ad hoc. Second, it provides a model-consistent revealed-productivity measure that improves on simple revealed comparative advantage indices. Third, it uses the estimated structure to ask a welfare question: how important is industry-level Ricardian comparative advantage for the gains from trade?

8.4 Implications for empirical trade

The paper implies that empirical tests of comparative advantage should pay close attention to the exact object predicted by theory. Total exports, net exports, and simple market shares may not correspond to the structural prediction of the model. Correcting for openness, using appropriate fixed effects, and recognizing the difference between observed and fundamental productivity are all essential for credible empirical work.

8.5 Implications for welfare and counterfactuals

The counterfactual results are especially important because they separate the existence of Ricardian comparative advantage from its welfare magnitude. Ricardian productivity differences clearly shape trade patterns, but removing industry-level productivity differences lowers average gains from trade by only a modest amount in the baseline. This does not mean productivity is unimportant; rather, it means that trade costs, demand differences, and within-industry heterogeneity also play major roles in sustaining trade and welfare gains.

8.6 Limitations

The analysis is limited by the aggregation level of the productivity data, which is roughly two-digit manufacturing. Ricardian forces could look more important at more disaggregated product levels. The IV strategy relies on R&D expenditure affecting trade through productivity. The counterfactual results depend on the model structure, including Frechet productivity draws, Cobb-Douglas upper-tier preferences, and the way absolute productivity is adjusted to hold wages fixed. The data are from 1997 and cover a relatively rich-country manufacturing sample.

8.7 Takeaway

The enduring takeaway is that Ricardo can be tested and quantified in a modern empirical framework. The paper shows that relative productivity differences strongly predict relative exports, but it also shows that the welfare contribution of industry-level Ricardian comparative advantage is smaller than the textbook intuition might suggest once realistic trade costs and demand heterogeneity are incorporated.